
MINI PROJECT – Natural Language Processing

Objective:

→ Choose a Natural Language Processing application based on modern deep learning techniques such as Transformers, Large Language Models (LLMs), or advanced sequence models. The project can be based on **text data, speech (voice) data, or a combination of both**. Students are free to choose their topic. Example applications (for inspiration) include:

- Question Answering systems
- Text summarization or simplification
- Paraphrase detection and semantic similarity
- Hate speech or toxicity detection
- Topic modeling
- Cross-lingual NLP tasks
- Speech recognition (Speech-to-Text)
- Emotion recognition from speech
- Spoken language understanding (speech + text)
- Simplified voice-based chatbot (speech input/output)

2. **Model Development**

Students should implement at least one NLP model and perform the necessary data preprocessing steps (cleaning, tokenization, etc.). Train and evaluate the model using appropriate metrics such as accuracy, F1-score, BLEU score, or others depending on the task. The use of modern architectures such as BERT, T5, or other Transformer-based models is strongly encouraged. Comparing multiple approaches or models is considered a valuable extension.

3. **Application Deployment**

Deploy the best-performing model in a simple application such as a web interface or an interactive script. The application should allow users to input text or speech (if applicable) and obtain predictions or generated outputs.

4. **Report and Submission**

Prepare a detailed report presenting the problem statement, the dataset(s) used, the methodology, the model architecture, the experimental results, the discussion, and the conclusion with possible improvements. Projects should be carried out in groups of up to 5 or 6 students. The code must be shared via a GitHub repository.

Consultation sessions will be organized at the end of the semester for each group to present their work, including the developed application and the results obtained on the used datasets, and to receive detailed feedback.